

HPC and AI Benchmarks on GPU, Interconnect, and Storage

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Outline

A bottom-up tour of the AICR cluster

- **Cluster & hardware overview** — B200, RTX 6000, IB, Vast
- **GPU compute** — gpu-fryer GEMM throughput
- **Interconnect** — NCCL collectives
- **End-to-end AI training** — Megatron-LM (with PyTorch)
- Storage — **multinode PyTorch I/O sweep** (dataloader, torch_save, DCP)
- **Key takeaways** and recommendations

AICR cluster — hardware at a glance

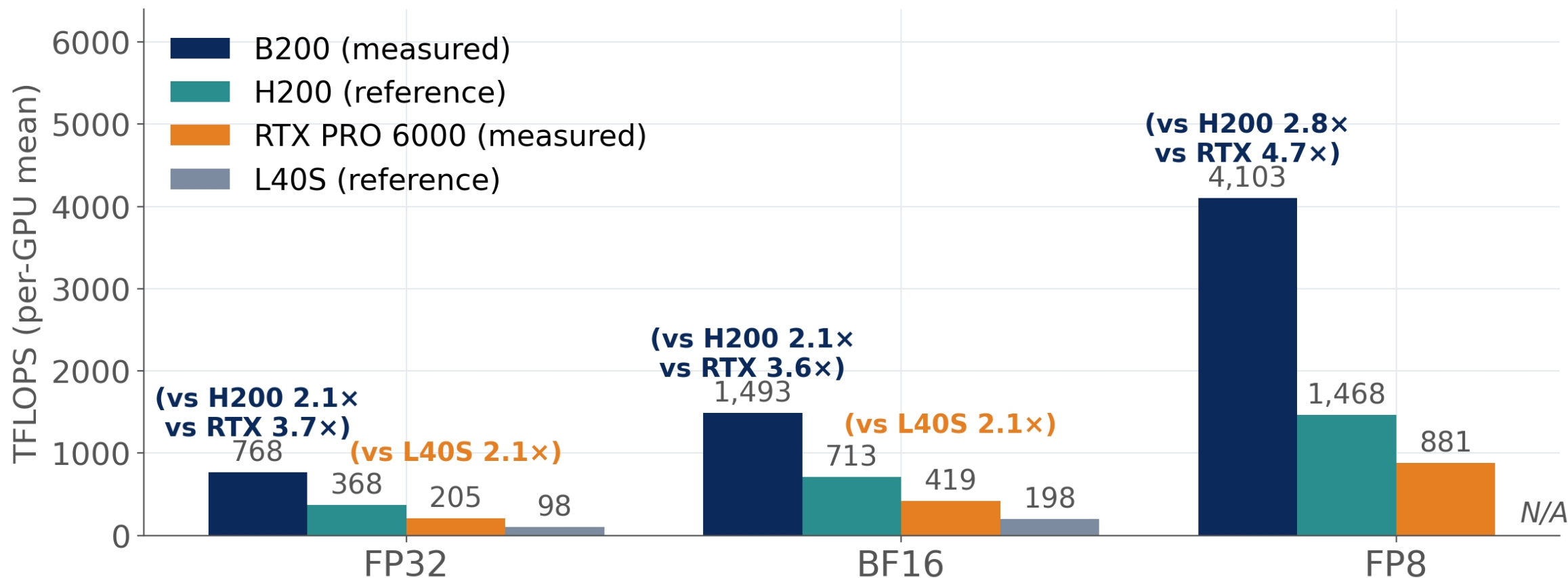
	B200 node	RTX PRO 6000 node
GPUs / node	8× NVIDIA B200	8× RTX PRO 6000 Blackwell SE
GPU memory	192 GB HBM3e	96 GB GDDR7
Intra-node interconnect	NVLink 5.0 + NVSwitch (~900 GB/s/GPU/dir)	PCIe Gen5 + Infinity Fabric only
NIC config	8× NDR 400 Gb/s (1 per GPU)	NDR IB (per-node)
CPU topology	2× AMD EPYC 9575F, NPS=4 (8 NUMA dies)	2× AMD EPYC 9575F, NPS=4 (8 NUMA dies)
Storage backend	Vast NFS over RDMA (Gen5 / Ceres 1350)	same

Fixed system issues

- **nvidia_peermem** loaded — enables GPUDirect RDMA over IB
- **ACS disabled, iommu=off** — restores PCIe P2P
- Applied to **both** B200 and RTX PRO 6000 nodes

GPU compute — gpu-fryer dense Tensor-Core GEMM

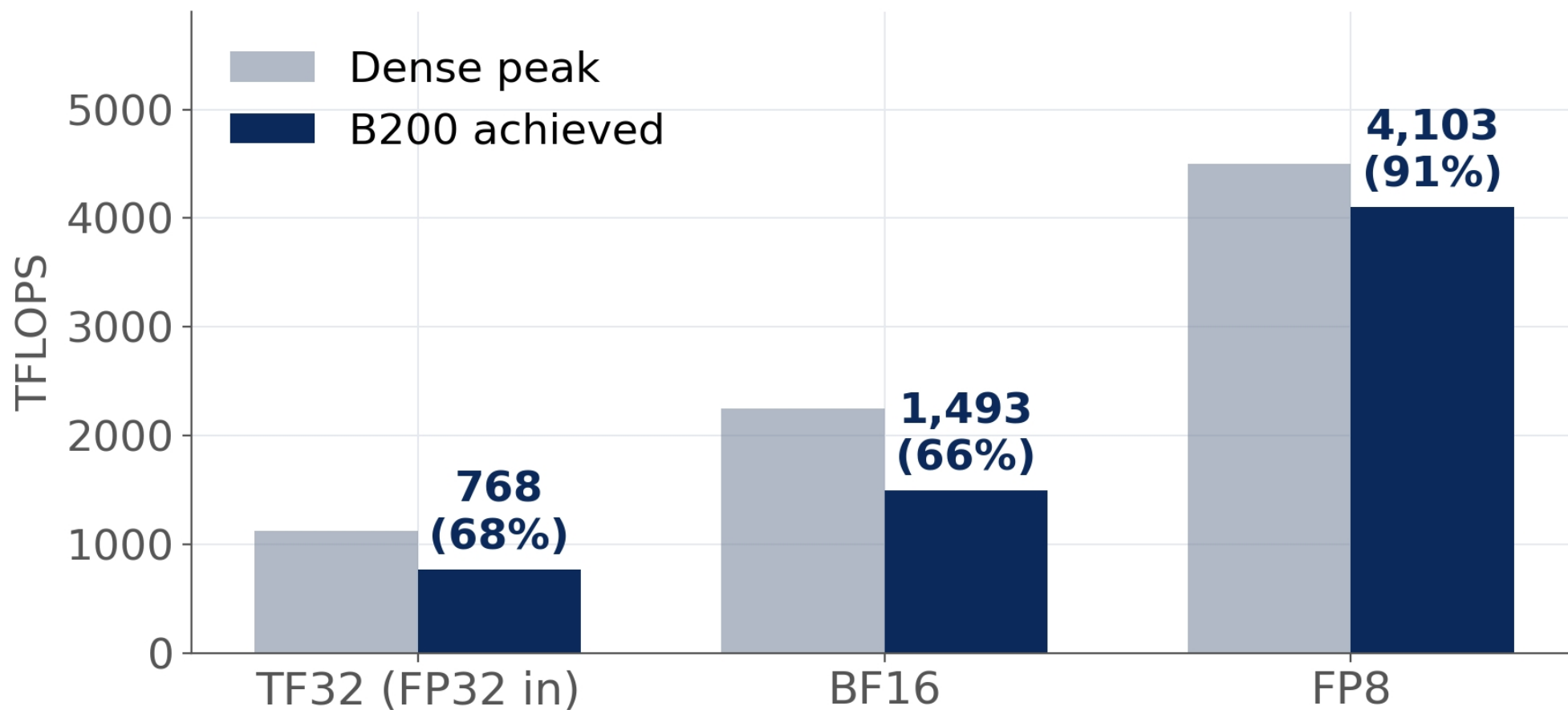
Per-GPU mean across 8 GPUs · gpu-fryer



- B200 $\approx$ **2.1x H200** (FP32/BF16), rising to **2.8x at FP8** — clearly optimized for low precision
- **BF16 for training. FP8 for inferencing.**
- B200 $\approx$ **3.6-4.7x RTX PRO 6000** (same arch, different die)
- All available nodes report "**All GPUs healthy**".

B200 — achieved TFLOPS vs dense peak

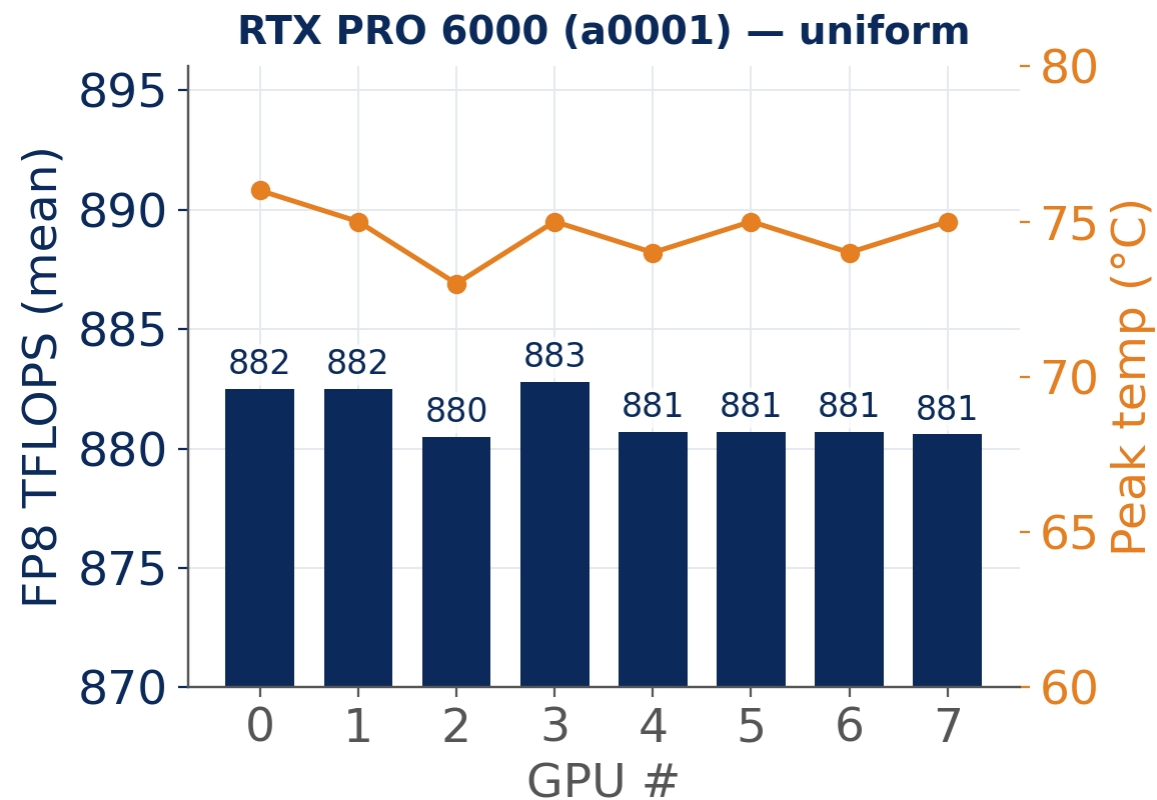
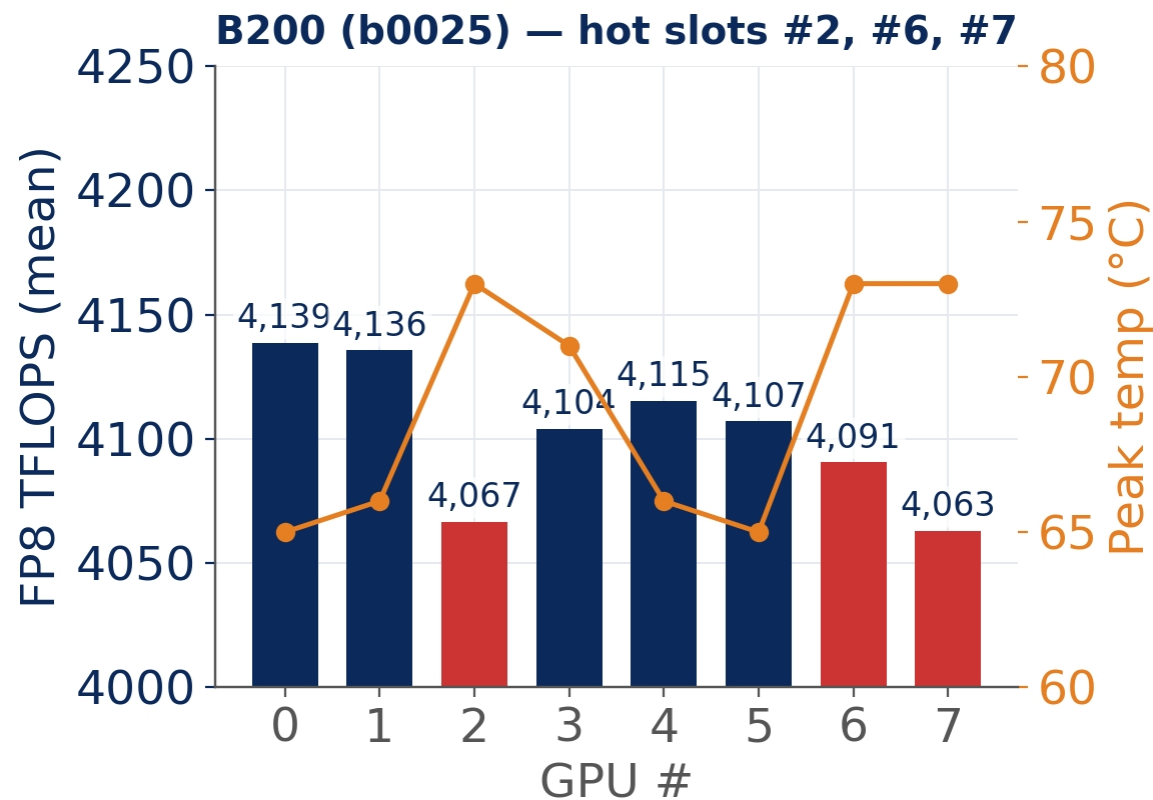
TF32 (FP32 input), BF16, FP8 · per-GPU mean



- FP8 is the standout — **91 % of dense peak**
- $\text{peak_FLOPS} = (\text{\#Tensor Cores}) \times (\text{ops per Tensor Core per cycle for that precision}) \times (\text{boost clock})$

Per-GPU uniformity and thermals

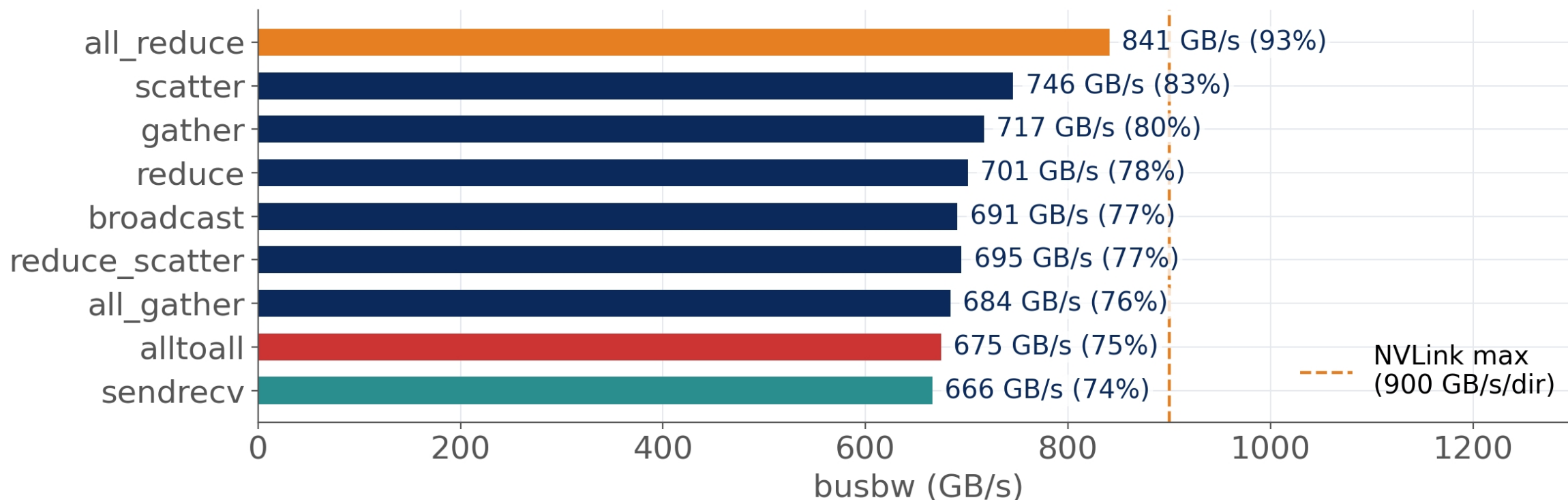
Healthy silicon, but B200 chassis shows three persistent hot slots



- TFLOPS are **uniform** across all GPUs on both B200 and RTX PRO 6000 nodes.
- Three B200 GPUs (#2, #6, #7) run **slightly warmer at 73 °C** — consistent with liquid-cooling topology, where coolant warms as it passes upstream GPUs before reaching downstream slots. All readings remain **within normal operating range**.

Interconnect — NCCL Intra-node, B200 (NVLink 5 / NVSwitch)

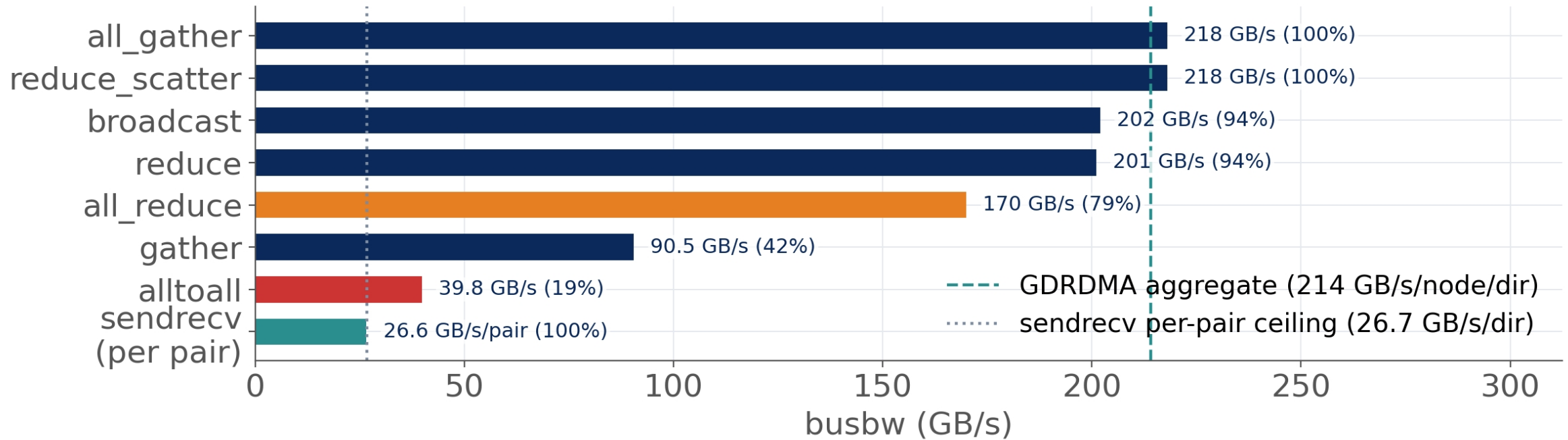
8 GPUs on b0027 · 16 GB message · NVLink max = 900 GB/s/dir



- All collectives achieve high bandwidth, enabled by the **NVLink 5 / NVSwitch** fabric.
- **AllReduce best at 93 %** (841 GB/s) — NCCL uses NVSwitch-optimal NVLS, not a ring; near-saturates the fabric.
- **SendRecv at 74 %** (666 GB/s) — NVSwitch P2P arbitration overhead; bidirectional traffic reduces peak unidirectional speed.
- **AllToAll at 75 %** (675 GB/s) — N^2 P2P algorithm overhead; lowest among symmetric collectives, but still healthy intra-node.

NCCL Inter-node, B200 (2 nodes × 8 GPUs, NDR IB)

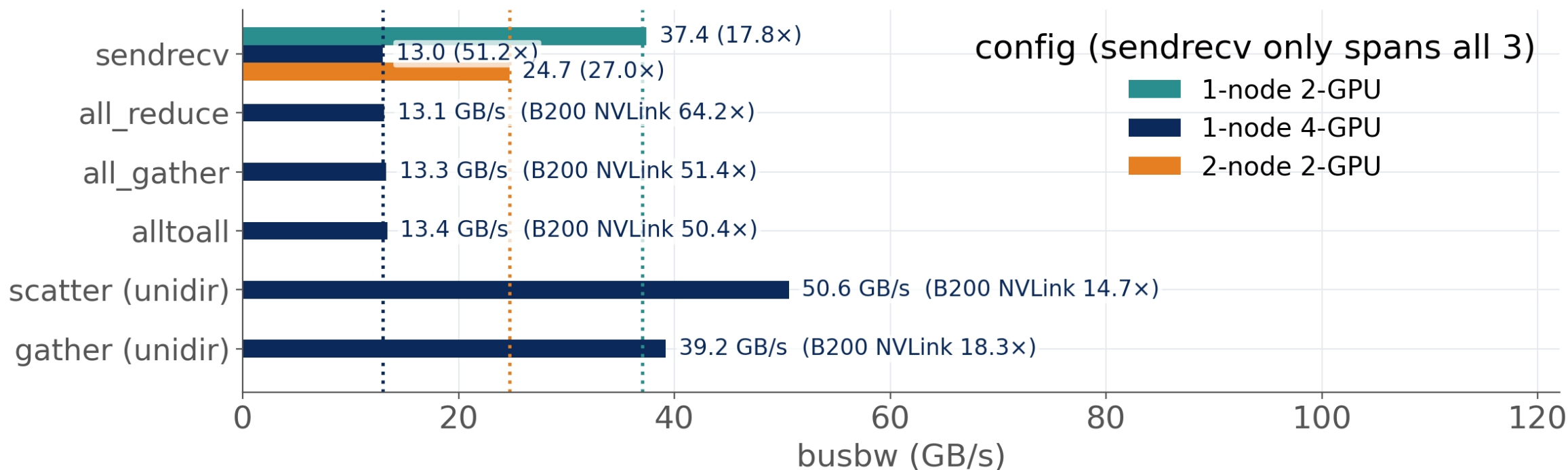
GDRDMA per-pair bidir ceiling = 26.7 GB/s/dir · aggregate ≈ 214 GB/s/node/dir



- **SendRecv at ~100 %** of the per-pair bidir ceiling (26.6 / 26.7 GB/s) — the GPU's PCIe DMA budget is shared between transmit and receive, halving the per-direction throughput vs the NDR unidir spec (50 GB/s). A **silicon-level constraint**; no NCCL tuning can lift it.
- **AllReduce** (170 GB/s) is good. Enabling SHARP could roughly double it to ≈ 340 GB/s. SHARP is in-network computing. The IB switch does the reduction.
- **AllToAll at 19 %** — NCCL ring-based N^2 P2P chunks are not pipelined across NICs; an **algorithm limitation**, not a hardware issue.

RTX PRO 6000 NCCL — no NVLink, PCIe is the bottleneck

NPS=4: every GPU pair traverses Infinity Fabric between NUMA dies · no GPU pair shares a PCIe switch · 16 GB message



- Three regimes, **three different bottlenecks** (dotted lines): **1-node 2-GPU** hits the GPU's PCIe DMA bidir budget (~37 GB/s); **1-node 4-GPU** saturates Infinity Fabric (~13 GB/s); **2-node 2-GPU** sendrecv hits the GDRDMA bidir DMA budget on the GPU (~24.7 GB/s). **1node 8-GPU** low speed at 0.04 GB/s — an issue to be fixed.
- The **2 → 4 GPU drop** happens because no GPU pair on this node shares a PCIe switch: every transfer must cross IF between NUMA dies, and once 4 pairs run concurrently the on-package fabric saturates.
- **vs B200 NVLink**: collectives are 15–64x slower on RTX PRO 6000; root-anchored unidir scatter/gather fare best because they avoid IF's bidir contention.
- Recommendation: **RTX PRO 6000 is best for single-GPU or 2-GPU workloads.**

Implications for parallel training (B200)

Mapping NCCL primitives to DL parallelism patterns

Parallelism	Primary NCCL op	1-node busbw	2-node busbw	Outlook
Data Parallel (DDP)	AllReduce	841 GB/s	170 GB/s	Inter-node-bound
Pipeline Parallel	SendRecv (P2P)	666 GB/s	26.6 GB/s	Inter-node at hard PCIe-DMA HW limit
Tensor Parallel	AR / AG / RS	684-841	170-218 GB/s	Keep within 1 node; cross-node ~4x slower
MoE (expert dispatch)	AllToAll	675 GB/s	39.8 GB/s	Inter-node severely bottlenecked (19% of agg.)

- All collectives perform well **intra-node**; **inter-node** performance varies by collective.
- **DDP scaling**: activate SHARP — gradient sync drops from 1032 ms to $\approx$ half.
- **PP across nodes**: 1 GB activation $\rightarrow$ $\sim$ 37 ms over 26.6 GB/s sendrecv. Plan pipeline bubbles accordingly.
- **TP** > 8 ranks must cross nodes — drops from 841 to 218 GB/s; **prefer staying within a node**.
- **MoE at scale**: constrain experts to single node, or use NVSHMEM-based / custom AllToAll.